Singular-value decomposition

AP

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Background

Spectral Methods: Traditional

Eigenpairs

$$A\vec{e} = \lambda\vec{e}$$

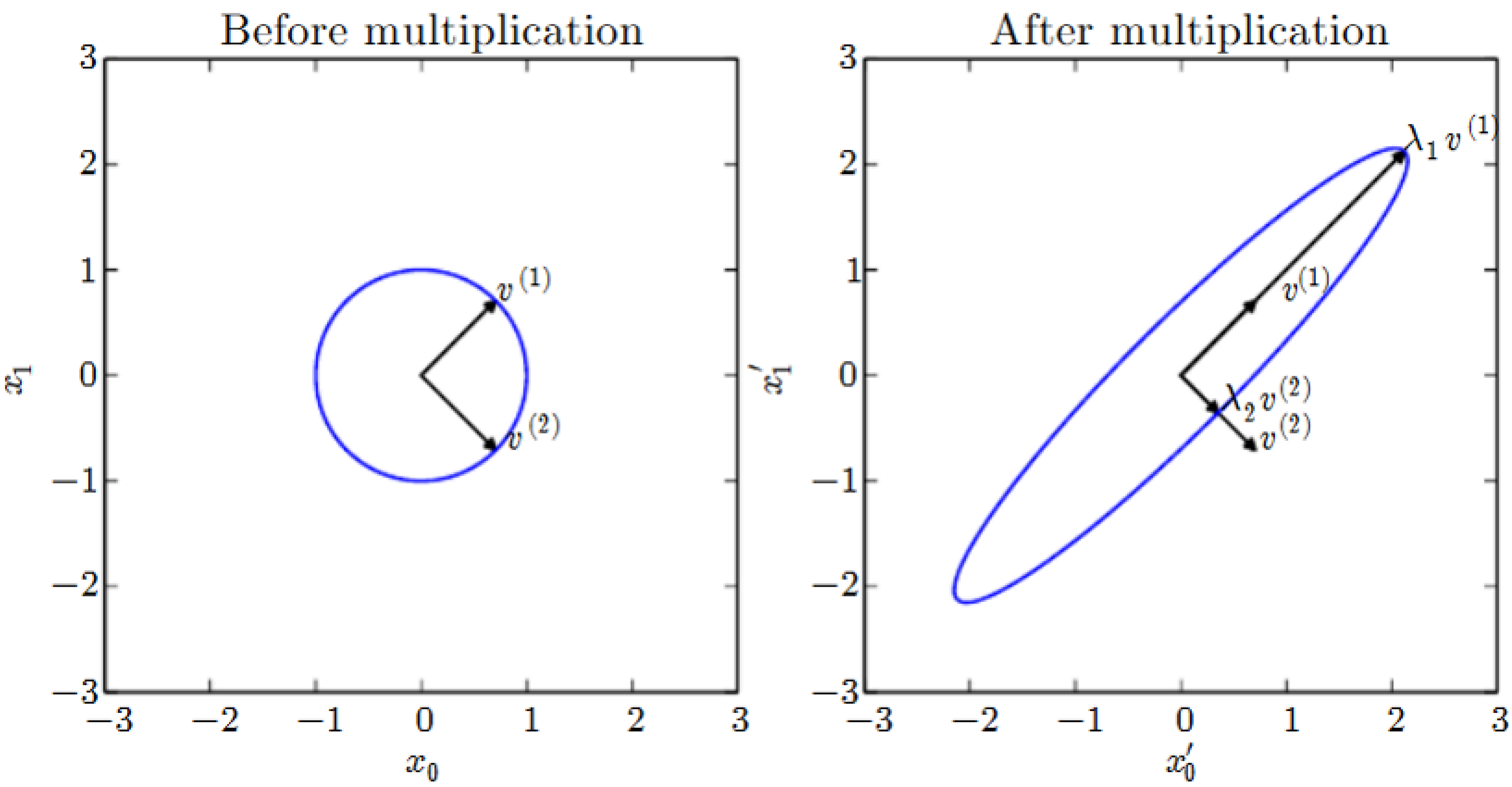

Figure 2.3: An example of the effect of eigenvectors and eigenvalues. Here, we have a matrix $\boldsymbol{A}$ with two orthonormal eigenvectors, $\boldsymbol{v}^{(1)}$ with eigenvalue λ_1 and $\boldsymbol{v}^{(2)}$ with eigenvalue λ_2. *(Left)*We plot the set of all unit vectors $\boldsymbol{u} \in \mathbb{R}^2$ as a unit circle. *(Right)*We plot the set of all points $\boldsymbol{A}\boldsymbol{u}$. By observing the way that $\boldsymbol{A}$ distorts the unit circle, we can see that it scales space in direction $\boldsymbol{v}^{(i)}$ by λ_i.

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Eigendecomposition:

$$M = Q\Lambda Q^{-1}$$

M is interpreted as the combination of three specific transformations
Computing its inverse, M^{-1}, is now much simpler

The importance of Matrix inversion

M. inversion solves linear systems of linear equations

$$M\vec{x} = \vec{v}$$

When M and $\vec{v}$ are know, we find $\vec{x}$

$$M^{-1}M\vec{x} = M^{-1}\vec{v}$$

$$I\vec{x} = M^{-1}\vec{v}$$

M. inversion underpins regression

The solution $\vec{x} = M^{-1}\vec{v}$ gives us the regression coefficients that define the linear regression model.

To estimate a new datapoint, say, $[a, b]$, just plug in the solution: $\hat{y} = [a, b] \cdot \vec{x}$

Regression seen as solving the underlying linear system by matrix inversion

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Regression seen as solving the underlying linear system by matrix inversion

Practical aspects

Textbook inversion takes $\Theta(n^3)$ operations: unfeasible

Advanced a. in Numpy run in $\Theta(n^2)$ (but check for numeric issues)

Compute M^{-1} once then store and re-use it for different $\vec{v}$

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Square matrix inversion

Thanks to properties of the Q matrix, where each column is an e-vector, inversion is simplified

$$M_{(m \times m)} = Q \Lambda Q^{-1}$$

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$$M^{-1} = (Q\Lambda Q^{-1})^{-1}$$

inverse factor order: $M^{-1} = (Q^{-1})^{-1}\Lambda^{-1}Q^{-1}$

$$M^{-1} = Q\Lambda^{-1}Q^{-1}$$

where Λ^{-1} is obtained by simply substituting each λ_i on the diagonal with $\frac{1}{\lambda_i}$.

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Singular-Value Decomposition

For rectangular matrices $A_{(m \times n)}$ eigendecomposition is not defined

Instead, SVD provides a similar decomposition of the data matrix

The procedure is more complex

Until 1955, no general procedure for inverting rectangular m. was available

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The data matrix

$A_{(m \times n)}$

There are m points (rows) inside a space of n dimensions (columns)

Each point is represented by an m-dimensional row vector

Each dimension is represented by an n-dimensional vector

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By multiplying a rectangular matrix by its transpose we obtain a square matrix that reprents an ‘internal’ relationship:

$$M_{(m \times m)} = A_{(m \times n)} \times A^T_{(n \times m)}$$

$$N_{(n \times n)} = A^T_{(n \times m)} \times A_{(m \times n)}$$

Let’s extract their respective eigenpairs.

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New matrix: U

$U_{(m\times m)}$: columns made up of eigenvectors of $M_{(m\times m)} = A_{(m\times n)} \times A^T_{(n\times m)}$

notice that e-vectors are always orthogonal with each other: $\vec{U_i}^T \cdot \vec{U_j} = 0 \; (i \neq j)$

this will simplify computation

However, some further orthogonal column will have to be introduced with external methods as only $min(m,n)$ non-zero eigenpairs can be obtained.

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New matrix: V

$V_{(n\times n)}$: colums made up of eigenvectors of $N_{(n\times n)}$

Again, e-vectors are orthogonal to each other: $\vec{V_i}^T \cdot \vec{V_j} = 0 \; (i \neq j)$

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New matrix: D (or Σ)

dispose the eigenvalues of N on the main diagonal of a rectangular m. that will be 0 everywhere else

$D_{(n\times m)}$ where $D_{ii} = \sigma_i$ are the singular values

$\sigma_i = \sqrt{\lambda_i}$ the i-th e-value of $N = A^T A$

(it can also be constructed with $M = AA^T$)

Finally…

$$A_{(m\times n)} = U_{(m\times m)} D_{(m\times n)} V^T_{(n\times n)}$$

- U is a orthogonal m. of left-singular (col.) vectors
- D is a diagonal matrix of singular values
- V is a orthogonal m. of right-singular (col.) vectors

Please see § 2.7 of [Goodfellow et al.]

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Consequences

SVD generalises eigen-decomposition:

- any real matrix has one
- even non-square matrices admit one

overconstrained systems (more rows than cols.) can now be solved

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Worked-out example

Example

$$A_{(3\times 2)}\vec{x} = \vec{v}$$

Focus on

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}$$

$$AA^T = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix} \begin{pmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{pmatrix} = \begin{pmatrix} 5 & 11 & 17 \\ 11 & 25 & 39 \\ 17 & 39 & 61 \end{pmatrix}$$

Extract the eigenpairs: $\lambda_1 = 91, v_1 = [0.15, 0.33, 0.52]$

Unfortunately, $\lambda_1 = \lambda_2 = 0$: we need a couple more orthogonal vectors to make up our matrix U

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$$A^T A = \begin{pmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix} = \begin{pmatrix} 35 & 44 \\ 44 & 56 \end{pmatrix}$$

The eigenvalues are $\lambda_1 = 90.54$ and $\lambda_2 = 0.26$
The singular values are $\sigma_1 = \sqrt(\lambda_1) = 9.54$ and $\sigma_2 = \sqrt(\lambda_2) = 0.51$

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$$D = \begin{pmatrix} 9.53 & 0 \\ 0 & 0.51 \\ 0 & 0 \end{pmatrix}$$

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Putting it all together:

$$A = \begin{pmatrix} 0.15 & 0.231 & 0.882 \\ 0.33 & 0.527 & 0.216 \\ 0.52 & 0.823 & -0.451 \end{pmatrix} \begin{pmatrix} 9.53 & 0 \\ 0 & 0.51 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0.62 & 0.79 \\ -0.79 & 0.62 \end{pmatrix}$$

Even though this formulation is far from the original and not easily interpreted, it has good properties.

Study plan

Background study

Ian Goodfellow, Yoshua Bengio and Aaron Courville: Deep Learning, MIT Press, 2016.

available in HTML and PDF; it is a refresher of notation and properties: no examples and no exercises.

It can be read in the background.

- Phase 1: read §§ 2.1—2.7, then § 2.11.
- Phase 2: read §§ 2.8—2.10

Moore-Penrose pseudo-inverse

Motivations

solve linear systems

$$A\vec{x} = \vec{v}$$

for non-square (rectangular) matrices:

- rows > columns: the problem is overconstrained (no solution?)
- rows < columns: the problem is overparametrized (infinite sols.?)

Reminder: the ideal procedure

If A is squared (m=n) and non-singular ($|A| \neq 0$) then

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$$I\vec{x} = A^{-1}\vec{v}$$

Compute A^{-1} once, run for different values of $\vec{v}$

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The pseudo-inverse

Given A_{(m n)}, Penrose seeks a matrix A^+ that would work the same as the left inverse:

$$A^+ A \approx I$$

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strenghten up the main diagonal before inversion:

$$A^+ = \lim_{\alpha \to 0}(A^T A + \alpha I)^{-1} A^T$$

Thanks to A^+, over-constrained linear systems can now be solved (w. approximation)

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SVD leads to approx. inversion

for the decomposition

$$A = UDV^T$$

$$A^+ = VD^+U^T$$

where D^+, such that $D^+D = I$ is easy to calculate: D is diagonal.

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Does $A^+A \approx I$?
Yes, because U and V are s. t. $U^T U = V V^T = I$.

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$$V D^+ U^T \cdot U D V^T \; =$$

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$$VD^+U^T \cdot UDV^T \ =$$

$$VD^+IDV^T \ =$$

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$$VIV^T = VV^T = I$$